

PREDICTIVE DISCRIMINATION UNDER THE RANDOM EFFECTS MODEL

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Summary: This paper investigates normal-theory predictive discrimination under the random effects model. Assuming that the elements within any randomly selected population are normally distributed with mean vector μ and common covariance matrix Σ , and that over different populations μ has a normal distribution with mean vector ξ and covariance matrix T , the predictive density of an observation, given a training sample from each of k randomly selected populations, is derived under the assumption that it has come from a particular population. Both the usual diffuse prior distribution for the unknown parameters Σ , ξ and T and Box and Tiao's non-informative prior distribution under the random effects model are considered. The results are applied to a stratigraphic problem in underground mining.

1. INTRODUCTION

The predictive Bayesian approach to Discriminant analysis has been eloquently described by Geisser (1964, 1966). Briefly, it may be summarised as follows:

Given a training sample of p -dimensional data vectors

$TS = \{x_{ij}; j=1, \dots, n_i; i=1, \dots, k\}$ from k populations $\pi_1, \pi_2, \dots, \pi_k$ and an observation x of unknown origin, this approach consists in evaluating the posterior probability, given TS and the underlying model together with any known parameters, that x belongs to π_r for $r=1$ to k , and then assigning x to that population for which this probability is the greatest. Following the exposition of Geisser (1966), suppose that each $\pi_r, r=1, \dots, k$ is specified by a probability density function $f(\cdot | \theta_r, \psi_r)$, where θ_r is the set of unknown parameters and ψ_r the set of known parameters, and that x has a prior probability q_i of belonging to π_i .

Let $\theta = \bigcup_{i=1}^k \theta_i$ and $\psi = \bigcup_{i=1}^k \psi_i$ be the sets of distinct unknown and known parameters, respectively, $q = \{q_1, \dots, q_k\}$ the set of prior probabilities and $g(\theta | \psi)$ the joint prior density of θ given ψ . The posterior probability that x comes from π_r is then:

$$(1) \quad P(x \in \pi_r | TS, \psi, q) = \frac{q_r f(x | TS, \psi, \pi_r)}{\sum_{i=1}^k q_i f(x | TS, \psi, \pi_i)}$$

where

$$(2) \quad f(x | TS, \psi, \pi_r) = \int_{\theta} f(x | \theta_r, \psi_r) p(\theta | TS, \psi) d\theta$$

is the predictive density of x assuming that it comes from π_r ,

$$(3) \quad p(\theta | TS, \psi) = g(\theta | \psi) L(TS | \theta, \psi)$$

is the joint posterior density of θ , given the training sample and ψ , and

$L(TS|\theta, \psi)$ is the joint likelihood of the training sample.

The situation discussed by Geisser (1964, 1966) is that in which π_i is characterised by a (multivariate) normal distribution and it is assumed that $\pi_1, \pi_2, \dots, \pi_k$ are fixed populations predetermined by the problem faced. This paper deals with the case where the populations π_i have themselves been randomly selected from some parent population in advance of the experiment. Once the k populations have been selected we are then faced with the more usual problem of classifying into one of the k (now fixed) populations. In different experiments there are different sets of populations π_i , in general with different numbers of populations k , all drawn independently from the same parent population.

More specifically, we assume that population π_i is characterised by a p -dimensional normal distribution $N_p(\mu_i, \Sigma)$, where μ_i has been randomly selected from a $N_p(\xi, T)$ distribution, in contrast to the "fixed effects" model where μ_i is assumed to have come from a noninformative prior distribution. The other parameter Σ and the two "hyper-parameters" ξ and T are, however, assumed to have a joint noninformative prior distribution.

In the next two sections expressions for the predictive density (2) will be derived under the random effects model, firstly using a generalisation of the usual diffuse prior distribution for Σ , ξ and T , and secondly using Box and Tiao's (1973) noninformative prior distribution under the random effects model. In both situations special attention will be given to the univariate case. The results obtained are expressed in terms of hypergeometric functions of matrix argument, and in section 4 the computational aspects of evaluating these posterior probabilities will be discussed. In the final section the results will be applied to a stratigraphic problem in underground mining.

2. DERIVATION OF THE PREDICTIVE DENSITY

We will assume that the unknown parameters Σ , ξ and T have the following general type of diffuse joint prior density function:

$$(4) \quad g(\Sigma, \xi, T) d\Sigma d\xi dT \propto |\Sigma|^{-\frac{1}{2}v_1} |T|^{-\frac{1}{2}v_2} d\Sigma d\xi dT$$

This prior has been used by Geisser (1964) and is more general than the usual noninformative or diffuse prior distribution discussed by Geisser and Cornfield (1963), for which $v_1 = v_2 = p+1$.

Therefore, given the training sample TS , consisting of n_i observations from π_i , $i=1, \dots, k$, the prior distribution of μ_i under the random effects model and the joint prior distribution of Σ , ξ and T given by (4), the predictive density of a new observation x , under the hypothesis $x \in \pi_r$, becomes:

$$f(x|TS, v_1, v_2, \pi_r) = \int \int_{\Sigma, \mu} f(x|\mu, \Sigma, \pi_r) p(\mu, \Sigma|TS) d\mu d\Sigma$$

where

μ is the $p \times k$ matrix $(\mu_1, \mu_2, \dots, \mu_k)$,

$$f(x|\mu, \Sigma, \pi_r) = (2\pi)^{-\frac{1}{2}p} |\Sigma|^{-\frac{1}{2}} \exp\{-\frac{1}{2}(x-\mu_r)' \Sigma^{-1}(x-\mu_r)\},$$

$p(\mu, \Sigma|TS) \propto g(\mu, \Sigma) L(TS|\mu, \Sigma)$ is the joint posterior density of μ and Σ , $L(TS|\mu, \Sigma)$ is the joint likelihood of the training sample,

$$g(\mu, \Sigma) \propto \int \int_T g(\mu|\xi, T) g(\Sigma, \xi, T) d\xi dT$$

and $g(\mu|\xi, T)$ is the joint distribution of $\mu_1, \mu_2, \dots, \mu_k$ under the random effects model.

Putting this all together and reparameterizing, for convenience, to Σ^{-1} and T^{-1} with corresponding Jacobian $J(\Sigma, T \rightarrow \Sigma^{-1}, T^{-1}) = |\Sigma^{-1} T^{-1}|^{-(p+1)}$ we get, after a little simplification:

$$f(x|TS, v_1, v_2, \pi_r) \propto \int_{\Sigma^{-1}} \int_{\mu} |\Sigma^{-1}|^{\frac{1}{2}(N+v_1-2p-2)} \\ \times \exp\{-\frac{1}{2} \sum_{i=1}^k \sum_{j=1}^{n_i^*} (x_{ij} - \mu_i)' \Sigma^{-1} (x_{ij} - \mu_i)\} \\ \times \int_{T^{-1}} \int_{\xi} |T^{-1}|^{\frac{1}{2}(k+v_2-2p-2)} \exp\{-\frac{1}{2} \sum_{i=1}^k (\mu_i - \xi)' T^{-1} (\mu_i - \xi)\} d\xi dT^{-1} d\mu d\Sigma^{-1}$$

where we have used the notation:

$$n_i^* = \begin{cases} n_i & i \neq r \\ n_{r+1} & i = r \end{cases},$$

$$x_{rn_{r+1}^*} = x$$

and

$$N = \sum_{i=1}^k n_i^* = \sum_{i=1}^k n_{i+1}.$$

Noting that:

$$\sum_{i=1}^k (\mu_i - \xi)' T^{-1} (\mu_i - \xi) = \text{tr}(T^{-1} A_{\mu}) + k(\xi - \mu_{\cdot})' T^{-1} (\xi - \mu_{\cdot})$$

$$\text{where } A_{\mu} = \sum_{i=1}^k (\mu_i - \mu_{\cdot})(\mu_i - \mu_{\cdot})'$$

$$\text{and } \mu_{\cdot} = \sum_{i=1}^k \mu_i, \quad .$$

the inner pair of integrals become:

$$\begin{aligned} & \int_{T^{-1}} |T^{-1}|^{\frac{1}{2}(k+v_2-2p-3)} \exp\{-\frac{1}{2}\text{tr}(T^{-1}A_\mu)\} \\ & \times \int_{\xi} |T^{-1}|^{\frac{1}{2}} \exp\{-\frac{1}{2}k(\xi-\mu.)'T^{-1}(\xi-\mu.)\} d\xi dT^{-1} \\ & \propto \int_{T^{-1}} |T^{-1}|^{\frac{1}{2}(k+v_2-2p-3)} \exp\{-\frac{1}{2}\text{tr}(T^{-1}A_\mu)\} dT^{-1} \end{aligned}$$

This integrand is proportional to the Wishart density function, the constant of proportionality being (see, for example, Press, 1972):

$$|A_\mu|^{\frac{1}{2}(k+v_2-p-2)} / \{2^{\frac{1}{2}p(k+v_2-p-2)} \Gamma_p(\frac{1}{2}(k+v_2-p-2))\}.$$

where $\Gamma_p(\frac{1}{2}n)$ is the multivariate gamma function. Using this result, and the expansion:

$$\sum_{i=1}^k \sum_{j=1}^{n_i^*} (x_{ij} - \mu_i)' \Sigma^{-1} (x_{ij} - \mu_i) = \text{tr}(\Sigma^{-1} A_2^*) + \sum_{i=1}^k n_i^* (\mu_i - x_{i.}^*)' \Sigma^{-1} (\mu_i - x_{i.}^*)$$

$$\text{where } A_2^* = \sum_{i=1}^k \sum_{j=1}^{n_i^*} (x_{ij} - x_{i.}^*)(x_{ij} - x_{i.}^*)'$$

$$\text{and } x_{i.}^* = \frac{1}{n_i^*} \sum_{j=1}^{n_i^*} x_{ij} = \begin{cases} x_{i.} & , \quad i \neq r \\ x_{r.} + \frac{x - x_{r.}}{n_r + 1} & , \quad i = r \end{cases}$$

the predictive density can be written:

$$f(x|TS, v_1, v_2, \pi_r) \propto \int_{\Sigma^{-1}} |\Sigma^{-1}|^{\frac{1}{2}(n+v_1-k-2p-2)} \exp\{-\frac{1}{2}\text{tr}(\Sigma^{-1}A_2^*)\} \\ \times \int_{\mu} |A_{\mu}|^{-\frac{1}{2}(k+v_2-p-2)} |\Sigma^{-1}|^{\frac{1}{2}k} \exp\{-\frac{1}{2} \sum_{i=1}^k n_i^* (\mu_i - x_{i.}^*)' \Sigma^{-1} (\mu_i - x_{i.}^*)\} d\mu d\Sigma^{-1}$$

The inner integral is proportional to the $-\frac{1}{2}(k+v_2-p-2)^{\text{th}}$ moment of the generalized variance of a random sample $\mu_1, \mu_2, \dots, \mu_k$ where the μ_i are independently distributed as $N(x_{i.}^*, \frac{1}{n_i^*}\Sigma)$. In order to be able to evaluate this expected value explicitly, we have to make the assumption that the n_i^* are all equal, say $n_i^* = n^*, \forall i$. Under this assumption, A_{μ} can be considered to have a p -dimensional noncentral Wishart distribution with $(k-1)$ degrees of freedom, parameter matrix $\frac{1}{n^*}\Sigma$ and noncentrality matrix:

$$\Omega^* = \frac{1}{2}\Sigma^{-1}A_1^*$$

$$\text{where } A_1^* = n^* \sum_{i=1}^k (x_{i.}^* - x_{..}^*)(x_{i.}^* - x_{..}^*)'$$

$$\text{and } x_{..}^* = \frac{1}{k} \sum_{i=1}^k x_{i.}^*.$$

So the inner integral is proportional to the $-\frac{1}{2}(k+v_2-p-2)^{\text{th}}$ moment of the generalized variance corresponding to the $W_p(k-1; \frac{1}{n^*}\Sigma; \Omega^*)$ distribution.

Constantine (1963) gives the following expression for the t -th moment of the generalized variance corresponding to the $W_p(\nu; \Sigma; \Omega)$ distribution:

$$\mu_t' = \frac{\Gamma_p(\frac{1}{2}\nu+t)}{\Gamma_p(\frac{1}{2}\nu)} |2\Sigma|^t \exp\{-\text{tr}\Omega\} {}_1F_1(\frac{1}{2}\nu+t; \frac{1}{2}\nu; \Omega) \\ \text{for } t > -\frac{1}{2}(\nu-p+1) \text{ and } \nu > p-1$$

where ${}_1F_1(a; b; \Omega)$ is the confluent hypergeometric function with matrix argument defined by James (1954).

Applying this result we get:

$$f(x|TS, v_1, v_2, \pi_r) \propto \int_{\Sigma^{-1}} |\Sigma^{-1}|^{\frac{1}{2}(N+v_1+v_2-3p-4)} \exp\{-\frac{1}{2}\text{tr}\Sigma^{-1}A_3^*\} \\ \times {}_1F_1(\frac{1}{2}(p+1-v_2); \frac{1}{2}(k-1); \frac{1}{2}\Sigma^{-1}A_1^*) d\Sigma^{-1} \\ \text{for } v_2 < 2 \text{ and } k > p$$

$$\text{where } A_3^* = A_1^* + A_2^* = \sum_{i=1}^k \sum_{j=1}^{n^*} (x_{ij} - x_{..}^*)(x_{ij} - x_{..}^*)' .$$

To evaluate this integral we use the expansion:

$${}_1F_1(a; b; \Omega) = \sum_{j=0}^{\infty} \sum_{x(j)} \frac{a^{\{x(j)\}}}{b^{\{x(j)\}}} c_{x(j)}(\Omega)/j!$$

where $x(j)$ is a partition of the integer j of weight p

of the form $\{j_1, j_2, \dots, j_p\}$ where $j_i \geq 0$ and $\sum_{i=1}^p j_i = j$,

$c_{x(j)}(\Omega)$ is the zonal polynomial in the eigenvalues of corresponding to partition $x(j)$,

$$a^{\{x(j)\}} = \prod_{j=1}^p (a - \frac{1}{2}(i-1))^{[j_i]}$$

$$\alpha^{[j]} = \alpha(\alpha+1) \dots (\alpha+j-1)$$

and $\sum_{x(j)}$ denotes the sum over all possible partitions

$x(j)$ of j .

Substituting this and interchanging the order of summation and integration yields

$$f(x|TS, v_1, v_2, \pi_r) \propto \sum_{j=0}^{\infty} \sum_{x(j)} \frac{(\frac{1}{2}(p+1-v_2))^{x(j)}}{(\frac{1}{2}(k-1))^{x(j)} j!} \int_{\Sigma^{-1}} |\Sigma^{-1}|^{\frac{1}{2}(N+v_1+v_2-3p-4)} \\ \times \exp\{-\frac{1}{2}\text{tr}\Sigma^{-1}A_3^*\} C_{x(j)}(\frac{1}{2}\Sigma^{-1}A_1^*) d\Sigma^{-1}$$

The integral may now be evaluated using the fundamental integral identity (Constantine, 1963)

$$\int_S \exp\{-\text{tr} RS\} |S|^{t-\frac{1}{2}(p+1)} C_{x(j)}(ST) dS \\ = \Gamma_p(t) t^{x(j)} C_{x(j)}(R^{-1}T) |R|^{-t}.$$

This yields:

$$(5) \quad f(x|TS, v_1, v_2, \pi_r) \propto |A_3^*|^{-\frac{1}{2}(N+v_1+v_2-2p-3)} \sum_{j=0}^{\infty} \sum_{\Sigma(j)} \frac{(\frac{1}{2}(p+1-v_2))^{x(j)}}{(\frac{1}{2}(k-1))^{x(j)} j!} \\ \times (\frac{1}{2}(N+v_1+v_2-2p-3))^{x(j)} C_{x(j)}(A_3^{*-1}A_1^*) / j! \\ = |A_3^*|^{-\frac{1}{2}(N+v_1+v_2-2p-3)} {}_2F_1(\frac{1}{2}(p+1-v_2), \frac{1}{2}(N+v_1+v_2-2p-3); \\ \frac{1}{2}(k-1); A_3^{*-1}A_1^*)$$

$$\text{for } v_2 < 2 \text{ and } k > p$$

where ${}_2F_1(a_1, a_2; b_1; \Omega)$ is the hypergeometric function with matrix argument defined by James (1954).

REMARKS

(a) The restriction that v_2 must be less than 2 for the predictive density to exist means that T cannot have the usual diffuse prior distribution with $v_2 = p+1$. It is rather surprising that the prior distribution of the second stage "hyper-parameter" T in the random effects model presents a problem, whereas that of the corresponding first stage parameter Σ presents none at all, at least within the framework of the diffuse prior distributions (4). Therefore, if we assign the values $v_1 = p+1$ and $v_2 = 1$, giving Σ the usual noninformative prior distribution relative to the likelihood function of the multivariate normal distribution, and T a prior distribution that is only very approximately so, then predictive density (5) becomes:

$$(6) \quad f(x|TS, \pi_r) \propto |A_3^*|^{-\frac{1}{2}(N-p-1)} {}_2F_1(\frac{1}{2}p, \frac{1}{2}(N-p-1); \frac{1}{2}(k-1); A_3^{*-1}A_1^*)$$

for $k > p$

(b) In the univariate case

$$x(j) = j, \quad c_{x(j)}(\omega) = \omega^j, \quad a\{x(j)\} = a[j]$$

and

$${}_2F_1(a_1, a_2; b_1; \omega) = \sum_{j=0}^{\infty} \frac{a_1^{[j]} a_2^{[j]}}{b_1^{[j]}} \frac{\omega^j}{j!}$$

so that the predictive density becomes (using the values $v_1 = 2$ and $v_2 = 1$):

$$(7) \quad f(x|TS, \pi_r) \propto A_3^{*-\frac{1}{2}(N-2)} \sum_{j=0}^{\infty} \frac{(\frac{1}{2})^{[j]} (\frac{1}{2}(N-2))^{[j]}}{(\frac{1}{2}(k-1))^{[j]}} \frac{(A_1^*/A_3^*)^j}{j!}$$

$$= A_3^{*-\frac{1}{2}(N-2)} F(\frac{1}{2}, \frac{1}{2}(N-2); \frac{1}{2}(k-1); A_1^*/A_3^*)$$

$$\text{where } A_1^* = n^* \sum_{i=1}^k (x_{i.}^* - x_{..}^*)^2, \quad A_3^* = \sum_{i=1}^k \sum_{j=1}^{n^*} (x_{ij} - x_{..}^*)^2$$

and $F(a_1, a_2; b_1; \omega)$ is the hypergeometric function.

- (c) The assumption that $n_i^* = n^*$, $\forall i = 1, \dots, k$, effectively implies that $n_i = n = n^*$ (which is commonly assumed in the analysis of random effects models, see, for example, Johnson and Leone (1964), p. 13) and that when the predictive density is evaluated for each π_r , $r=1, \dots, k$ in turn, one of the observations x_{rj} is chosen from the training sample from π_r and is replaced by x , the observation to be classified. Thus the effective total sample size of the training sample becomes

$\sum_{i=1}^k n_i - 1$, and $N = \sum_{i=1}^k n_i$. A sensible rule for choosing which observation x_{rj} to replace by x is to choose that x_{rj} which is closest to the sample mean from π_r , as measured by the Mahalanobis distance, i.e. choose x_{rj} that minimizes

$$d_r^2(x_{rj}) = (x_{rj} - x_r)' S^{-1} (x_{rj} - x_r)$$

The following computational formulae are useful when computing $f(x|TS, \pi_r)$ in turn for each of the populations π_r , $r=1, \dots, k$:

$$A_1^* = A_1 + (x - x_{rj})(x_r - x_{..})' + (x_r - x_{..})(x - x_{rj})' + \frac{k-1}{kn} (x - x_{rj})(x - x_{rj})'$$

$$A_2^* = A_2 - \frac{1}{n} (x - x_{rj})(x - x_{rj})' - (x_{rj} - x_r)(x_{rj} - x_r)' + (x - x_r)(x - x_r)'$$

and $A_3^* = A_1^* + A_2^*$

where $A_1 = n \sum_{i=1}^k (x_i - x_{..})(x_i - x_{..})'$

and $A_2 = \sum_{i=1}^k \sum_{j=1}^n (x_{ij} - x_i)(x_{ij} - x_i)'$

are the between groups and within groups sums of squares matrices, respectively, for the training sample.

3. BOX AND TIAO'S PRIOR DISTRIBUTION

Box and Tiao (1973) point out that under the random effects model with equal sample sizes from each group, the sampling theory estimator $\hat{T}$ of T :

$$\hat{T} = \frac{1}{n}(S_1 - S_2)$$

$$\text{where } S_1 = \frac{1}{k-1} A_1 \quad \text{and} \quad S_2 = \frac{1}{k(n-1)} A_2,$$

may not be positive definite. In order to avoid this possibility within the Bayesian framework, they propose the following non-informative joint prior density for the parameters Σ , ξ and T :

$$(8) \quad g(\Sigma, \xi, T) d\Sigma d\xi dT \propto |\Sigma|^{-\frac{1}{2}(p+1)} |\Sigma + nT|^{-\frac{1}{2}(p+1)} d\Sigma d\xi dT$$

This prior distribution suffers from the drawback that the within-groups sample size n appears in its density function, making it dependent not only on the form of, but also on the actual likelihood function itself. However, as shall be shown, use of this prior distribution does avoid the problem, encountered in the previous section, regarding the existence of the predictive density under the random effects model.

Applying this prior distribution, with v_1 and v_2 replacing the parameter $(p+1)$ in the first and second positions in expression (8), and using the same notation as in the previous section, we obtain the following integral expression for the predictive density, given the training sample and the assumption that $x \in \pi_r$:

$$\begin{aligned} f(x|TS, v_1, v_2, \pi_r) \propto & \int \int \int |\Sigma|^{-\frac{1}{2}N} \exp\left\{-\frac{1}{2} \sum_{i=1}^k \sum_{j=1}^n (x_{ij} - \mu_i)' \Sigma^{-1} (x_{ij} - \mu_i)\right\} \\ & \times |T|^{-\frac{1}{2}k} \exp\left\{-\frac{1}{2} \sum_{i=1}^k (\mu_i - \xi)' T^{-1} (\mu_i - \xi)\right\} |\Sigma|^{-\frac{1}{2}v_1} |\Sigma + nT|^{-\frac{1}{2}v_2} d\mu d\xi d\Sigma dT \end{aligned}$$

where, as before, it has been assumed that $n_i^* = n$, $\forall i$ and that

the j -th observation x_{rj} from π_r has been replaced by x , and x has been re-labelled x_{rj} .

This integral is evaluated in the appendix, yielding the expression:

$$(9) \quad f(x|TS, v_1, v_2, \pi_r) \propto |A_3^*|^{-\frac{1}{2}(N+v_1+v_2-2p-3)} {}_2F_1\left(\frac{1}{2}(p+1), \frac{1}{2}(N+v_1+v_2-2p-3); \frac{1}{2}(k+v_2-1); A_3^{*-1}A_1^*\right)$$

$$\text{for } N + v_1 + v_2 - 2 > 3p \text{ and } v_2 + k - 1 > 2p$$

Notice that, in contrast to the previous section where the parameter v_2 was restricted to being less than 2 for the predictive density to exist, v_1 and v_2 may now both assume the value $p+1$ as long as N and k are both greater than p . Therefore, substituting the values $v_1 = v_2 = p+1$ in (9) we get:

$$(10) \quad f(x|TS, \pi_r) \propto A_3^{*-\frac{1}{2}(N-1)} {}_2F_1\left(\frac{1}{2}(p+1), \frac{1}{2}(N-1); \frac{1}{2}(k+p); A_3^{*-1}A_1^*\right)$$

In the univariate case this becomes:

$$(11) \quad f(x|TS, \pi_r) \propto A_3^{*-\frac{1}{2}(N-1)} F\left(1, \frac{1}{2}(N-1); \frac{1}{2}(k+1); A_1^*/A_3^*\right).$$

4. ON EVALUATING THE PREDICTIVE DENSITY

In order to try and evaluate the hypergeometric functions with matrix argument appearing in expressions (6) and (10) for the predictive density under the two different prior distributions, the suite of FORTRAN programs of van der Westhuyzen and Nagel (1979) for computing the zonal polynomials in the eigenvalues of a matrix Ω , corresponding to all the partitions of an integer j , were used. Because of limitations in computer time and space, a maximum value of $j = 18$ could be used. Unfortunately, in all the samples considered, the number of terms required for

the hypergeometric functions in either expression to converge was far in excess of this figure.

In the univariate case the hypergeometric functions are all expressed in terms of elementary functions and could readily be evaluated on a computer. The number of terms required for convergence (to an accuracy of 10^{-8}) was typically in the region of 300, which suggests that the number of terms required in the multivariate case will be quite beyond the capabilities of the programs of van der Westhuyzen and Nagel (1979).

5. A PRACTICAL APPLICATION

In this section the results derived earlier are applied to a stratigraphic problem in underground mining. In the Witwatersrand gold fields the gold bearing reef is one band - the pay band - in a sedimentary succession, and is usually visually unrecognisable. In badly faulted areas this pay band may fault away, and the miner then wishes to know the position in the sedimentary succession of the blank band facing him, from which he can deduce the new position of the pay band.

One method of identification is via the trace element geochemistry of the different bands. It is reasonable to suppose that the concentrations of trace elements within a particular band is characterised by a multivariate statistical distribution. This distribution changes as we move from band to band, reflecting the different geological conditions under which they were first laid down. As these different geological conditions may themselves be presumed to have been selected by some random mechanism, we have the situation where trace element concentrations drawn from different bands follow a random effects model.

In table 1, five observations of the concentrations of Cobalt, after suitable transformation (Fatti, 1979), from each of 15 bands in the Witwatersrand Reef are given. The corresponding Cobalt

concentrations from two samples of unknown origin are also given. Using the Shapiro-Wilk (1965) and Bartlett's tests, respectively, the data were found to satisfy the requirements of within-group and between-group normality as well as homoscedasticity between groups.

In table 2 the posterior probabilities under the random effects model corresponding to each of the fifteen bands are given for each of the two unknowns, assuming *a priori* that they are equally likely to have come from any of these bands. For comparison, the posterior probabilities under both forms of diffuse prior distributions, (7) and (11) respectively, are presented, as well as are those under the "fixed effects" model (Geisser, 1964, expression 2.4). The Mahalanobis distances between the unknowns and each of the fifteen bands are also given in table 2, giving the classifications under the classical approach to discriminant analysis (Anderson, 1958) for comparison.

Using this latter approach, unknown 1 would unhesitatingly be classified into the first band, the nearest contender, band 10, having a distance nearly 20 times greater. On the other hand, although the first band has the greatest posterior probability under the "fixed effects" model, that of band 10 is only fractionally lower.

The posterior probabilities under the random effects model are very little affected by the form of prior distribution used for the parameters Σ , ξ and T , be it the slightly modified form of Geisser and Cornfield's (1963) prior distribution considered in §2 or Box and Tiao's (1973) considered in §3. In contrast to the situation under the "fixed effects" model, band 11 now has the highest posterior probability with band 2 a close second.

In practice, however, using the predictive Bayesian approach under either the fixed or random effects models, unknown 1 would be considered unclassifiable because in neither case do the

posterior probabilities indicate a clear choice between the rock bands. The divergence between the classical and predictive Bayesian approaches observed here is in line with the findings of Aitchison, Habbema and Kay (1977), whose general conclusion is that the classical or "estimative" approach tends to give too optimistic a picture of the reliability of sample-based discrimination procedures.

The situation is far clearer with unknown 2. All four classification rules come out strongly in favour of either populations 13 or 14, the predictive approach under the random effects model using either form of noninformative prior distribution giving slight preference to band 14, whereas the other two marginally favour band 13.

The reason for the improved reliability of classification in the latter case is quite evident under the random effects model. Since unknown 2 is much further from the estimated mean, $\xi = 0,1464$, of the individual population means μ_i than unknown 1, one would expect better classification with it as the groups would tend to be much less clustered in its vicinity than they would be nearer to ξ .

It is interesting to note that the standard deviations of the sets of posterior probabilities under the random effects model are slightly smaller, using either form of prior distribution for the parameters Σ , ξ and T , than they are under the fixed effects model, indicating a greater degree of diffuseness of the posterior probabilities under the former model. This clearly results from the fact that the noninformative prior distribution used by Geisser (1964) for the population means under the fixed effects model implies, *a priori*, that the means are far more likely to be highly separated than in the case of the random effects model. (See also Cox and Hinkey (1974), p. 379).

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TABLE 1

CONCENTRATIONS OF COBALT (AFTER TRANSFORMATION)
FROM EACH OF 15 SEDIMENTARY BANDS

<u>Band No.</u>	<u>1</u>	<u>2</u>	<u>3</u>
	0,3858	0,4255	-0,1660
	0,5065	0,4008	0,3160
	0,4081	0,4735	0,1448
	0,3210	-0,3862	0,1572
	-0,2393	0,0569	-0,1533

<u>Band No.</u>	<u>4</u>	<u>5</u>	<u>6</u>
	0,6285	0,5217	0,3330
	0,3091	0,5099	0,6624
	0,2866	0,5490	0,5272
	0,3784	0,2981	0,1279
	0,2984	0,3222	0,3033

<u>Band No.</u>	<u>7</u>	<u>8</u>	<u>9</u>
	0,4148	-0,2589	-0,0765
	0,8251	-0,2214	-0,0939
	0,4799	0,0087	0,2947
	0,2183	0,0340	0,4301
	0,5873	-0,1673	-0,1776

<u>Band No.</u>	<u>10</u>	<u>11</u>	<u>12</u>
	0,5880	0,4872	0,1739
	0,5295	-0,0089	0,0322
	0,4256	0,5219	-0,5402
	-0,1759	-0,0709	-0,4637
	0,2849	0,0478	0,4625

<u>Band No.</u>	<u>13</u>	<u>14</u>	<u>15</u>
	-0,3224	-0,1766	0,2866
	-0,5506	-0,4704	0,3875
	-0,5330	-0,4110	0,3904
	-0,3700	-0,5465	0,3823
	-0,3114	-0,3710	0,2989

Grand mean: 0,1464

Unknown 1: 0,2865

Unknown 2: -0,4075

TABLE 2

POSTERIOR PROBABILITIES OF THE TWO UNKNOWN UNDER THE
RANDOM EFFECTS AND FIXED EFFECTS MODELS, AS WELL AS
THE CORRESPONDING MAHALANOBIS DISTANCES

Unknown 1

Band no.	Random Effects Model		Fixed effects model	Mahalanobis distances
	Diffuse prior	Box and Tiao's prior		
1	0,0844	0,0842	0,0955	0,0017
2	0,0947	0,0942	0,0899	0,1461
3	0,0598	0,0596	0,0660	0,8800
4	0,0810	0,0810	0,0897	0,1506
5	0,0876	0,0880	0,0806	0,4049
6	0,0793	0,0791	0,0884	0,1863
7	0,0681	0,0685	0,0677	0,8187
8	0,0371	0,0374	0,0294	2,8426
9	0,0883	0,0882	0,0693	0,7630
10	0,0822	0,0819	0,0943	0,0331
11	0,0978	0,0974	0,0900	0,1419
12	0,0388	0,0388	0,0391	2,1401
13	0,0068	0,0070	0,0032	8,4861
14	0,0093	0,0096	0,0039	7,9550
15	0,0849	0,0849	0,0929	0,0673
Standard deviation				
	0,0299	0,0297	0,0324	

Unknown 2

1	0,0161	0,0161	0,0109	8,0108
2	0,0232	0,0232	0,0219	6,1979
3	0,0648	0,0643	0,0580	3,7388
4	0,0068	0,0069	0,0041	10,6259
5	0,0056	0,0057	0,0022	12,3063
6	0,0056	0,0057	0,0037	10,9124
7	0,0025	0,0025	0,0011	14,2623
8	0,1294	0,1287	0,1506	1,4059
9	0,0522	0,0521	0,0524	3,9928
10	0,0090	0,0091	0,0066	9,3248
11	0,0232	0,0232	0,0217	6,2250
12	0,1190	0,1176	0,1185	1,9847
13	0,2623	0,2631	0,2715	0,0017
14	0,2709	0,2725	0,2714	0,0026
15	0,0094	0,0095	0,0055	9,8043
Standard deviation				
	0,0906	0,0909	0,0944	

APPENDIX

Evaluation of the integral in §3:

The exponent can be written as:

$$-\frac{1}{2} \left[\sum_{i=1}^k \sum_{j=1}^{n_i^*} (x_{ij} - x_{i.}^*)' \Sigma^{-1} (x_{ij} - x_{i.}^*) \right. \\ \left. + \sum_{i=1}^k \left\{ (\mu_i - x_{i.}^*)' (n_i^{*-1} \Sigma)^{-1} (\mu_i - x_{i.}^*) + (\mu_i - \xi)' T^{-1} (\mu_i - \xi) \right\} \right]$$

where n_i^* and $x_{i.}^*$ have been defined in the same way as in §2. Assuming now that $n_i^* = n v_i$ and using the identity given by Box and Tiao (1973) in their appendix A7.1 to simplify the second summation term yields:

$$-\frac{1}{2} \left[\text{tr} \Sigma^{-1} A_2^* + \sum_{i=1}^k (\mu_i - c_i)' (n \Sigma^{-1} + T^{-1}) (\mu_i - c_i) \right. \\ \left. + (\xi - x_{..}^*)' (n^{-1} \Sigma + T)^{-1} (\xi - x_{..}^*) \right] \\ = -\frac{1}{2} \left[\text{tr} \Sigma^{-1} A_2^* + \sum_{i=1}^k (\mu_i - c_i)' (n \Sigma^{-1} + T^{-1}) (\mu_i - c_i) \right. \\ \left. + \text{tr} (\Sigma + nT)^{-1} A_1^* + nk (\xi - x_{..}^*)' (\Sigma + nT)^{-1} (\xi - x_{..}^*) \right]$$

where A_1^* and A_2^* are defined §2

and $c_i = (n \Sigma^{-1} + T^{-1}) (n \Sigma^{-1} x_{i.}^* + T \xi)$

The integral now becomes:

$$f(x|TS, v_1, v_2, \pi_r) \propto \int_{\Sigma} \int_T |\Sigma|^{-\frac{1}{2}(N+v_1)} |\Sigma + nT|^{-\frac{1}{2}v_2} |T|^{-\frac{1}{2}k} \\ \times \exp\{-\frac{1}{2} \text{tr} (\Sigma^{-1} A_2^* + (\Sigma + nT)^{-1} A_1^*)\} \\ \times \int_{\xi} \exp\{-\frac{1}{2} nk (\xi - x_{..}^*)' (\Sigma + nT)^{-1} (\xi - x_{..}^*)\}$$

$$\times \int_{\mu} \exp\left\{-\frac{1}{2} \sum_{i=1}^k (\mu_i - c_i)' (n\Sigma^{-1} + T^{-1}) (\mu_i - c_i)\right\} d\mu d\xi dT d\Sigma$$

The inner two integrals are proportional

$$\text{to: } |\Sigma + nT|^{\frac{1}{2}} |n\Sigma^{-1} + T^{-1}|^{-\frac{1}{2}k},$$

$$\begin{aligned} \text{so: } f(x|TS, v_1, v_2, \pi_r) &\propto \int_{\Sigma} \int_T |\Sigma|^{-\frac{1}{2}(N+v_1-k)} |\Sigma+nT|^{-\frac{1}{2}(v_2+k-1)} \\ &\times \exp\left\{-\frac{1}{2} \text{tr}(\Sigma^{-1}A_2^* + (\Sigma+nT)^{-1}A_1^*)\right\} dT d\Sigma \end{aligned}$$

Making the transformation $Y = \Sigma$ and $Z = \Sigma + nT$ we get

$$\begin{aligned} f(x|TS, v_1, v_2, \pi_r) &\propto \int_Y |Y|^{-\frac{1}{2}(N+v_1-k)} \exp\left\{-\frac{1}{2} \text{tr} Y^{-1} A_2^*\right\} \\ &\times \int_{Z>Y} |Z|^{-\frac{1}{2}(v_2+k-1)} \exp\left\{-\frac{1}{2} \text{tr} Z^{-1} A_1^*\right\} dZ dY \end{aligned}$$

To evaluate the inner integral, we make the transformation $W = Z^{-1}$ with Jacobian $J(Z \rightarrow W) = |W|^{-(p+1)}$. Denoting this integral by I , we therefore get:

$$\begin{aligned} I &= \int_{W<Y}^{-1} |W|^{\frac{1}{2}(v_2+k-p-2)} \exp\left\{-\frac{1}{2} \text{tr} A_1^* W\right\} dW \\ &= \frac{\Gamma_p\left(\frac{1}{2}(v_2+k-p-2)\right) \Gamma_p\left(\frac{1}{2}(p+1)\right)}{\Gamma_p\left(\frac{1}{2}(v_2+k-1)\right)} |Y|^{-\frac{1}{2}(v_2+k-p-2)} \\ &\times {}_1F_1\left(\frac{1}{2}(v_2+k-p-2); \frac{1}{2}(v_2+k-1); -\frac{1}{2} Y^{-1} A_1^*\right) \end{aligned}$$

for $v_2+k-p-2 > p-1$, using Constantine's (1963) expression for the multivariate incomplete gamma function.

Applying the Kummer transformation formula (Constantine, 1963, p. 1279):

$${}_1F_1(a; b; \Omega) = \exp(\text{tr} \Omega) {}_1F_1(b-a; b; -\Omega)$$

and substituting back into the first integral, yields:

$$f(x|TS, v_1, v_2, \pi_r) \propto \int_Y |Y|^{-\frac{1}{2}(N+v_1+v_2-p-2)} \exp\{-\frac{1}{2}\text{tr} Y^{-1} A_3^*\} \\ \times {}_1F_1(\frac{1}{2}(p+1); \frac{1}{2}(v_2+k-1); \frac{1}{2}Y^{-1}A_1^*) dY$$

$$\text{where } A_3^* = A_1^* + A_2^*$$

$$= \sum_{j=0}^{\infty} \sum_{x(j)} \frac{(\frac{1}{2}(p+1))\{x(j)\}}{(\frac{1}{2}(v_2+k-1))\{x(j)\}j!} \int |Y|^{-\frac{1}{2}(N+v_1+v_2-p-2)} \\ \times \exp\{-\frac{1}{2}\text{tr} Y^{-1} A_3^*\} C_{x(j)}(\frac{1}{2}Y^{-1}A_1^*) dY .$$

Making the transformation $X = Y^{-1}$ and using the integral identity of Constantine (1963, p. 1271) finally yields:

$$f(x|TS, v_1, v_2, \pi_r) \propto \sum_{j=0}^{\infty} \sum_{x(j)} \frac{(\frac{1}{2}(p+1))\{x(j)\}}{(\frac{1}{2}(v_2+k-1))\{x(j)\}j!} \Gamma_p(\frac{1}{2}(N+v_1+v_2-2p-3)) \\ \times (\frac{1}{2}(N+v_1+v_2-2p-3))\{x(j)\} |\frac{1}{2}A_3^*|^{-\frac{1}{2}(N+v_1+v_2-2p-3)} C_{x(j)}(A_3^{*-1}A_1^*) \\ \text{for } N+v_1+v_2-2p-3 > p-1 \\ \propto |A_3^*|^{\frac{1}{2}(N+v_1+v_2-2p-3)} {}_2F_1(\frac{1}{2}(p+1), \frac{1}{2}(N+v_1+v_2-2p-3); \frac{1}{2}(v_2+k-1); A_3^{*-1}A_1^*) .$$